Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

3-Plane Localizer - 3-Plane Localizer

IMAGING PARAMETERS	
Imaging Mode	*2D*
Pulse Sequence	*Spin Echo*
Imaging Options	*Seq, EDR, TRF, Fast, SS, ARC*
Phase	*2.00*

SCAN RANGE	
FOV	*30.0*
Slice Thickness	*10.00*
Slice Spacing	*5.0*
Overlap Locations	*0*

ACQ TIMING	
Freq	*288*
Phase	*128*
Freq DIR	*Unswap*
# of Acq. Before Pause	*0*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

FMRI	
PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT	
Tag Type	*None*

TRICKS	
Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

SCAN TIMING	
TE	*80.0*
Number of Echoes	*1*
TR	*Minimum*
Receiver Bandwidth	*83.33*

IMAGE ENHANCE	
Filter Choice	*None*

USER CVS	
User CV Mask2	*0*

MULTI-PHASE	
Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION	
Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST	
Contrast Yes/No	*No*

3-Plane Localizer - 3-Plane Localizer

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

PET Task

PATIENT POSITION	
Patient Entry	*Head First*
Patient Position	*Supine*

SCAN SETUP	
Scan Description	*PET BRAIN Static 1HR*
Scan Type	*Static*
Gating Mode	*Off*
Cardiac State	*Off*
Start Location	*S 137.259*
End Location	*I 107.381*
Overlap	*0 Slices*
MRAC	*Yes*
Phantom AC Map	*None*
Nuclide	*11C*
Tracer	*[11C]PIB*
Number of PET Beds	*1*

CARDIAC/RESPIRATORY	
Binning Type	*NA*
Trigger Rejection	*NA*
Avg Trig/min	*NA*
% Dev. Allowed	*NA*

RECON2	
Recon Enabled	*Yes*
Series Description	*PET BRAIN MAC TOF STATIC Reg*
DFOV	*30.000 cm*
Recon Method	*TOF*
Cardiac 3D Filter	*No*
Recon Start Location	*S 137.259*
Recon End Location	*I 107.381*
R/L Center	*0.000 mm*
A/P Center	*18.974 mm*
Z-Axis Filter	*Standard*
Filter Cutoff	*6.400 mm*
Subsets	*28*
Iterations	*6*
Matrix Size	*256 x 256*
Truncation Completion	*Yes*
Q.Static	*No*
Sharp IR	*No*
Beds Reconstructed	*1*
Attenuation Correction	*Yes*
Q.Clear Beta	*350.000*

PATIENT POSITION - REPLAY1	
Patient Entry	*Head First*
Patient Position	*Supine*

SCAN TIMING	
Pre-Scan Delay	*0*
Scan Time	*4200 sec*
Start on Count Rate	*0*
Stop on kcounts	*0*

SCAN DETAILS	
PET Bed 1 Scan Time	*300 sec*
PET Bed 2 Scan Time	*20 sec*
PET Bed 3 Scan Time	*60 sec*
PET Bed 4 Scan Time	*300 sec*

RECON1	
Recon Enabled	*Yes*
Series Description	*PET BRAIN NAC TOF STATIC Reg*
DFOV	*30.000 cm*
Recon Method	*TOF*
Cardiac 3D Filter	*No*
Recon Start Location	*S 137.259*
Recon End Location	*I 107.381*
R/L Center	*8.831 mm*
A/P Center	*7.474 mm*
Z-Axis Filter	*Standard*
Filter Cutoff	*6.400 mm*
Subsets	*28*
Iterations	*6*
Matrix Size	*256 x 256*
Truncation Completion	*Yes*
Q.Static	*No*
Sharp IR	*No*
Beds Reconstructed	*1*
Attenuation Correction	*No*
Q.Clear Beta	*350.000*

PET Task

Property of Wallace Tumor Institute

SCAN SETUP - REPLAY1	
Scan Description	*DYNAMIC PET BRAIN C-11*
Scan Type	*Dynamic*
Gating Mode	*Off*
Cardiac State	*Off*
Start Location	*S 137.259*
End Location	*I 107.381*
Overlap	*0 Slices*
MRAC	*Yes*
Phantom AC Map	*None*
Nuclide	*[11C]PIB*
Tracer	*11C*
Number of PET Beds	*1*

RECON1 - REPLAY1	
Recon Enabled	*Yes*
Series Description	*DYNAMIC PET BRAIN MAC TOF REG*
DFOV	*30.000 cm*
Recon Method	*TOF*
Cardiac 3D Filter	*No*
Recon Start Location	*S 137.259*
Recon End Location	*I 107.381*
R/L Center	*2.517 mm*
A/P Center	*17.617 mm*
Z-Axis Filter	*Standard*
Filter Cutoff	*6.400 mm*
Subsets	*28*
Iterations	*6*
Matrix Size	*256 x 256*
Truncation Completion	*Yes*
Q.Static	*No*
Sharp IR	*No*
Beds Reconstructed	*1*
Attenuation Correction	*Yes*
Q.Clear Beta	*350.000*

RECON3 - REPLAY1	
Recon Enabled	*No*

SCAN TIMING - REPLAY2	
Pre-Scan Delay	*2400*
Scan Time	*1800 sec*
Start on Count Rate	*0*
Stop on kcounts	*0*

CARDIAC/RESPIRATORY - REPLAY2	
Binning Type	*NA*
Trigger Rejection	*NA*
Avg Trig/min	*NA*
% Dev. Allowed	*NA*

RECON2 - REPLAY2	
Recon Enabled	*No*

RECON3	
Recon Enabled	*Yes*
Series Description	*PET BRAIN MAC ZTE Static Reg*
DFOV	*60.000 cm*
Recon Method	*TOF*
Cardiac 3D Filter	*No*
Recon Start Location	*S 137.259*
Recon End Location	*I 107.381*
R/L Center	*0.000 mm*
A/P Center	*0.000 mm*
Z-Axis Filter	*Standard*
Filter Cutoff	*6.400 mm*
Subsets	*28*
Iterations	*6*
Matrix Size	*256 x 256*
Truncation Completion	*Yes*
Q.Static	*No*
Sharp IR	*No*
Beds Reconstructed	*1*
Attenuation Correction	*Yes*
Q.Clear Beta	*350.000*

SCAN TIMING - REPLAY1	
Pre-Scan Delay	*0*
Scan Time	*4200 sec*
Start on Count Rate	*0*
Stop on kcounts	*0*

CARDIAC/RESPIRATORY - REPLAY1	
Binning Type	*NA*
Trigger Rejection	*NA*
Avg Trig/min	*NA*
% Dev. Allowed	*NA*

RECON2 - REPLAY1	
Recon Enabled	*No*

PATIENT POSITION - REPLAY2	
Patient Entry	*Head First*
Patient Position	*Supine*

SCAN SETUP - REPLAY2	
Scan Description	*DYNAMIC PET BRAIN C-11*
Scan Type	*Dynamic*
Gating Mode	*Off*
Cardiac State	*Off*
Start Location	*S 137.259*
End Location	*I 107.381*
Overlap	*0 Slices*
MRAC	*Yes*
Phantom AC Map	*None*
Nuclide	*[11C]PIB*
Tracer	*11C*
Number of PET Beds	*1*

Property of Wallace Tumor Institute

RECON1 - REPLAY2	
Recon Enabled	*Yes*
Series Description	*Amyloid_PIB 40-70m ZTE*
DFOV	*25.600 cm*
Recon Method	*VuePoint*
Cardiac 3D Filter	*No*
Recon Start Location	*S 137.259*
Recon End Location	*I 107.381*
R/L Center	*2.517 mm*
A/P Center	*17.617 mm*
Z-Axis Filter	*Off*
Filter Cutoff	*6.400 mm*
Subsets	*16*
Iterations	*4*
Matrix Size	*192 x 192*
Truncation Completion	*Yes*
Q.Static	*No*
Sharp IR	*No*
Beds Reconstructed	*1*
Attenuation Correction	*Yes*
Q.Clear Beta	*350.000*
RECON3 - REPLAY2	
Recon Enabled	*No*

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

SIMULTANEOUS MR - Bed 1

MRAC 1 - PET BRAIN Static 1HR

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*MRAC 1 - PET BRAIN Static 1HR*

SCAN TIMING

Flip Angle	*5.0*
TE	*Min Full*
Number of Echoes	*2*
Receiver Bandwidth	*166.67*

IMAGE ENHANCE

Filter Choice	*None*

USER CVS

User CV6	*1.00*
User CV30	*1.00*
User CV34	*70.00*
User CV Mask2	*8*

MULTI-PHASE

Seperate Series	*0*
Trigger Delay without AV	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*LAVA*
Imaging Options	*EDR, Fast, Flex*
IDEAL	*15*

SCAN RANGE

FOV	*50.0*
Slice Thickness	*2.60*
Location per Slab	*120*
Overlap Locations	*0*
Number of Slices	*1*

ACQ TIMING

Freq	*256*
Phase	*128*
Freq DIR	*R/L*
NEX	*1.00*
Phase FOV	*0.75*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

MRAC 1 - PET BRAIN Static 1HR

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

SIMULTANEOUS MR - Bed 1

ZTE-MRAC GEM HNU 1 - PET BRAIN Static 1HR

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*ZTE-MRAC GEM HNU 1 - PET BRAIN Static 1HR*

SCAN TIMING

Flip Angle	*0.8*
Number of Echoes	*1*
Receiver Bandwidth	*62.50*
Recovery Time	*0*

IMAGE ENHANCE

Filter Choice	*None*

GATING/TRIGGER

Pause After Navigator Prescan	*0*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*Gradient Echo*
Imaging Options	*EDR, Fast, SqP*
PSD Name	*3dradial*

SCAN RANGE

FOV	*26.4*
Slice Thickness	*2.40*
Overlap Locations	*0*
Number of Slices	*1*

ACQ TIMING

Freq	*110*
Freq DIR	*R/L*
NEX	*4.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*
Spokes Per Segment	*512*

USER CVS

User CV Mask2	*0*

MULTI-PHASE

Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

ZTE-MRAC GEM HNU 1 - PET BRAIN Static 1HR

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

SIMULTANEOUS MR - Bed 1

SAG 3D FSPGR

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*SAG 3D FSPGR*

SCAN TIMING

Flip Angle	*11.0*
TE	*Min Full*
Number of Echoes	*1*
TI	*900*
Receiver Bandwidth	*31.25*

IMAGE ENHANCE

Filter Choice	*None*

USER CVS

User CV6	*1.00*
User CV30	*1.00*
User CV34	*100.00*
User CV Mask2	*520*

MULTI-PHASE

Seperate Series	*0*
Trigger Delay without AV	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
Multi b-values	*1000.0;*
Multi NEX Values	*1.0;*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*SPGR*
Imaging Options	*EDR, Fast, Asset, IrP*
Phase	*2.00*

SCAN RANGE

FOV	*25.6*
Slice Thickness	*1.00*
Location per Slab	*200*
Overlap Locations	*0*
Number of Slices	*1*

ACQ TIMING

Freq	*256*
Phase	*256*
Freq DIR	*S/I*
Fat Shift DIR	*Normal (S)*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*On*
Phase Correction	*No*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

FMRI

Initial State	*Control*
PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

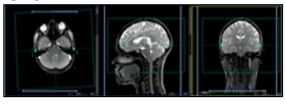
TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*2*

OTHERS

Protocol Notes

General Considerations:
PLEASE DO NOT OBLIQUE. DO NOT REDUCE NUMBER OF SLICES. DO NOT ADJUST PROTOCOL VALUES.
-3D imaging can improve the slice plane resolution compared to 2D imaging
-Prescribe the slices as shown



SAG 3D FSPGR

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

SIMULTANEOUS MR - Bed 1

SAG 3D T2 FLAIR

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*SAG 3D T2 FLAIR*

SCAN TIMING

TE	*108.0*
Number of Echoes	*1*
TR	*5000.0*
TI	*1563*
Echo Train Length	*182*
Receiver Bandwidth	*31.25*

IMAGE ENHANCE

Filter Choice	*None*

USER CVS

User CV5	*1.00*
User CV11	*20.00*
User CV30	*1.00*
User CV32	*1.00*
User CV34	*100.00*
User CV Mask2	*522*

MULTI-PHASE

Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
Multi b-values	*1000.0;*
Multi NEX Values	*1.0;*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*Cube T2 FLAIR*
Imaging Options	*EDR, Fast, ARC, IrP*
Phase	*2.00*
Slice	*2.00*

SCAN RANGE

FOV	*25.6*
Slice Thickness	*1.00*
Location per Slab	*200*
Overlap Locations	*0*
Number of Slices	*1*

ACQ TIMING

Freq	*256*
Phase	*256*
Freq DIR	*S/I*
Fat Shift DIR	*Normal (S)*
NEX	*1.00*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Quadrature (CP)*
Excitation Mode	*Selective*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*
Fat/Water Saturation	*Fat*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*2*

SAG 3D T2 FLAIR

Property of Wallace Tumor Institute

OTHERS	
Protocol Notes	*Cube 3D Imaging Considerations:* *PLEASE DO NOT OBLIQUE. PLEASE DO NOT REDUCE NUMBER OF SLICES. PLEASE DO NOT ADJUST PROTOCOL PARAMETERS.* *-Use Cube to acquire high resolution imaging in all three dimensions with the goal of acquiring isotropic voxels (all voxel dimensions are equl-height, width, and depth). Consider this information when modifying Cube T2 or Cube T2 FLAIR scan parameters.* *-Scan selections: 3D Mode, Fast Spine Echo family, Cube or Cube T2 FLAIR pulse, Extended Dynamic Range and IR Prepared Imaging Options.* *-Typically use Cube to acquire brain, C-spine, pelvis and keee images.* *-Cube brain scans acquired with User CV Cube Enhance set to 3 (Brain T1) can produce 3D isotropic images with reduced artifacts in comparison to a traditional 2D Spin Echo scan. However, residual flow artifacts and ringing artifact may still be present.* *-When acquring a Cube pelvis scan, it is recommended you acquire a sagittal or coronal plane. Acquiring an axial plane may result in annefact artifact.* *-From the scan parameters Details screen, consider selecting Excitation Mode: Non-Selective.* *-From the scan parameters Details screen, consider selecting Excitation Mode: Focus to redue phase alias artifact when a small FOV is used.*

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

SIMULTANEOUS MR - Bed 1

3D Ax SWAN

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*3D Ax SWAN*

SCAN TIMING

Flip Angle	*15.0*
TE	*22.5*
Number of Echoes	*11*
TR	*91.8*
Receiver Bandwidth	*83.33*

IMAGE ENHANCE

Filter Choice	*A*

USER CVS

User CV16	*6.00*
User CV26	*1.00*
User CV34	*70.00*
User CV Mask2	*520*

MULTI-PHASE

Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
Multi b-values	*1000.0;*
Multi NEX Values	*1.0;*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*SWAN*
Imaging Options	*FC, EDR, Fast, Asset*
Phase	*2.00*

SCAN RANGE

FOV	*25.6*
Slice Thickness	*1.50*
Location per Slab	*96*
Overlap Locations	*0*
Number of Slices	*2*

ACQ TIMING

Freq	*256*
Phase	*256*
Freq DIR	*R/L*
# of Acq. Before Pause	*0*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

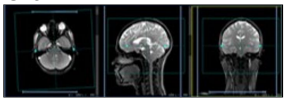
Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*2*

OTHERS

Protocol Notes	*General Considerations: -3D imaging can improve the slice plane resolution compared to 2D imaging -Prescribe the slices as shown*



3D Ax SWAN

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

SIMULTANEOUS MR - Bed 1

Multishell DTI ADNI

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*Multishell DTI ADNI*

SCAN TIMING

TE	*110.0*
Number of Echoes	*1*
TR	*9764.0*
Number of Shots	*1*

IMAGE ENHANCE

Filter Choice	*None*

USER CVS

User CV5	*1.00*
User CV11	*1000.00*
User CV Mask2	*0*

MULTI-PHASE

Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Optimized TE	*Yes*
Diffusion Directions	*Tensor*
Number of Diffusion Directions	*126*
Number of T2 Images	*3*
Dual Spin Echo	*Off*
Diffusion Tenser Processing Output	*No Selection*
Recon All Images	*Off*
Multi b-values	*2000.0;*
Multi NEX Values	*1.0;*
Real Time Field Adjustment	*1*

CONTRAST

Contrast Yes/No	*No*

IMAGING PARAMETERS

Imaging Mode	*2D*
Pulse Sequence	*Spin Echo*
Imaging Options	*EDR, Cla, EPI, DIFF, ARC, HB*
PSD Name	*epi2as*
Phase	*2.00*
HyperBand Slice	*2*

SCAN RANGE

FOV	*23.2*
Slice Thickness	*2.00*
Slice Spacing	*0.0*
Overlap Locations	*0*
Number of Slices	*70*

ACQ TIMING

Freq	*116*
Phase	*116*
Freq DIR	*R/L*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*Yes*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

OTHERS

Protocol Notes	*Important Tip: Hyperband DWI option uses a reverse polarity phase encoding technique which will show images slightly different from standard DWI without hyperband. If using standard polarity for hyperband DWI use the following psd type-in: epi2altoff*

Multishell DTI ADNI

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

SIMULTANEOUS MR - Bed 1

Multishell DTI ADNI

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*Multishell DTI ADNI*

SCAN TIMING

TE	*110.0*
Number of Echoes	*1*
TR	*9764.0*
Number of Shots	*1*

IMAGE ENHANCE

Filter Choice	*None*

USER CVS

User CV5	*1.00*
User CV11	*1000.00*
User CV Mask2	*0*

MULTI-PHASE

Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Optimized TE	*Yes*
Diffusion Directions	*Tensor*
Number of Diffusion Directions	*6*
Number of T2 Images	*3*
Dual Spin Echo	*Off*
Diffusion Tenser Processing Output	*No Selection*
Recon All Images	*Off*
Multi b-values	*1000.0;*
Multi NEX Values	*1.0;*
Real Time Field Adjustment	*1*

CONTRAST

Contrast Yes/No	*No*

IMAGING PARAMETERS

Imaging Mode	*2D*
Pulse Sequence	*Spin Echo*
Imaging Options	*EDR, Cla, EPI, DIFF, ARC, HB*
PSD Name	*epi2asaltoff*
Phase	*2.00*
HyperBand Slice	*2*

SCAN RANGE

FOV	*23.2*
Slice Thickness	*2.00*
Slice Spacing	*0.0*
Overlap Locations	*0*
Number of Slices	*70*

ACQ TIMING

Freq	*116*
Phase	*116*
Freq DIR	*R/L*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*Yes*
RF Drive Mode	*Preset*
Excitation Mode	*Selective*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

OTHERS

Protocol Notes	*Important Tip: Hyperband DWI option uses a reverse polarity phase encoding technique which will show images slightly different from standard DWI without hyperband. If using standard polarity for hyperband DWI use the following psd type-in: epi2altoff*

Multishell DTI ADNI

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

SIMULTANEOUS MR - Bed 1

HighResHippo

PET Task	*PET Task*
Bed	*Bed 1*
Series Description	*HighResHippo*

SCAN TIMING

Flip Angle	*122.0*
TE	*50.0*
Number of Echoes	*1*
TR	*3888.0*
Echo Train Length	*15*
Receiver Bandwidth	*20.83*

IMAGE ENHANCE

Filter Choice	*None*

USER CVS

User CV29	*1.00*
User CV30	*1.00*
User CV Mask2	*640*

MULTI-PHASE

Seperate Series	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

IMAGING PARAMETERS

Imaging Mode	*2D*
Pulse Sequence	*FSE-XL*
Imaging Options	*Fast, Asset, FR*
Phase	*2.00*

SCAN RANGE

FOV	*17.5*
Slice Thickness	*2.00*
Slice Spacing	*Intleave*
Overlap Locations	*0*
Number of Slices	*30*

ACQ TIMING

Freq	*448*
Phase	*448*
Freq DIR	*S/I*
Fat Shift DIR	*Normal (I)*
NEX	*1.00*
# of Acq. Before Pause	*0*
Phase FOV	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Quadrature (CP)*
Excitation Mode	*Selective*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*2*

HighResHippo

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

pCASL 1025

PET Task	*PET Task*
Bed	*pCASL 1025*
Series Description	*pCASL 1025*

SCAN TIMING

Number of Echoes	*1*
Receiver Bandwidth	*62.50*
Post Label Delay	*1025.0*

IMAGE ENHANCE

Filter Choice	*None*

GATING/TRIGGER

Pause After Navigator Prescan	*0*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*
Fat/Water Saturation	*Fat*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

OTHERS

Protocol Notes	*ASL Considerations: BOTTOM OF SLAB SHOULD BE AT BOTTOM OF CEREBELLUM. SUBJECTS EYES SHOULD BE OPEN. DO NOT OBLIQUE.* *ASL Scan Parameter Tradeoffs:*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*3DASL*
Imaging Options	*EDR, Fast, Spiral*

SCAN RANGE

FOV	*24.0*
Slice Thickness	*4.00*
Overlap Locations	*0*
Number of Slices	*1*

ACQ TIMING

Freq	*768*
Phase	*6*
Freq DIR	*R/L*
NEX	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Quadrature (CP)*
Excitation Mode	*Selective*

USER CVS

User CV2	*1.00*
User CV3	*1.00*
User CV4	*1.00*
User CV5	*1800.00*
User CV Mask2	*0*

MULTI-PHASE

Seperate Series	*0*
Delay after Acquisition without AV	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

pCASL 1025

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

pCASL 1525

PET Task	*PET Task*
Bed	*pCASL 1525*
Series Description	*pCASL 1525*
SCAN TIMING	
Number of Echoes	*1*
Receiver Bandwidth	*62.50*
Post Label Delay	*1525.0*
IMAGE ENHANCE	
Filter Choice	*None*
GATING/TRIGGER	
Pause After Navigator Prescan	*0*
FMRI	
PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*
SAT	
Tag Type	*None*
Fat/Water Saturation	*Fat*
TRICKS	
Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*
OTHERS	
Protocol Notes	*ASL Considerations: BOTTOM OF SLAB SHOULD BE AT BOTTOM OF CEREBELLUM. SUBJECTS EYES SHOULD BE OPEN. DO NOT OBLIQUE.* *ASL Scan Parameter Tradeoffs:*

IMAGING PARAMETERS	
Imaging Mode	*3D*
Pulse Sequence	*3DASL*
Imaging Options	*EDR, Fast, Spiral*
SCAN RANGE	
FOV	*24.0*
Slice Thickness	*4.00*
Overlap Locations	*0*
Number of Slices	*1*
ACQ TIMING	
Freq	*768*
Phase	*6*
Freq DIR	*R/L*
NEX	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Quadrature (CP)*
Excitation Mode	*Selective*
USER CVS	
User CV2	*1.00*
User CV3	*1.00*
User CV4	*1.00*
User CV5	*1800.00*
User CV Mask2	*0*
MULTI-PHASE	
Seperate Series	*0*
Delay after Acquisition without AV	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*
DIFFUSION	
Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*
CONTRAST	
Contrast Yes/No	*No*

pCASL 1525

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

pCASL 2025

PET Task	*PET Task*
Bed	*pCASL 2025*
Series Description	*pCASL 2025*

SCAN TIMING

Number of Echoes	*1*
Receiver Bandwidth	*62.50*
Post Label Delay	*2025.0*

IMAGE ENHANCE

Filter Choice	*None*

GATING/TRIGGER

Pause After Navigator Prescan	*0*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*
Fat/Water Saturation	*Fat*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

OTHERS

Protocol Notes	*. ASL Considerations: BOTTOM OF SLAB SHOULD BE AT BOTTOM OF CEREBELLUM. SUBJECTS EYES SHOULD BE OPEN. DO NOT OBLIQUE* *ASL Scan Parameter Tradeoffs:*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*3DASL*
Imaging Options	*EDR, Fast, Spiral*

SCAN RANGE

FOV	*24.0*
Slice Thickness	*4.00*
Overlap Locations	*0*
Number of Slices	*1*

ACQ TIMING

Freq	*768*
Phase	*6*
Freq DIR	*R/L*
NEX	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Quadrature (CP)*
Excitation Mode	*Selective*

USER CVS

User CV2	*1.00*
User CV3	*1.00*
User CV4	*1.00*
User CV5	*1800.00*
User CV Mask2	*0*

MULTI-PHASE

Seperate Series	*0*
Delay after Acquisition without AV	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

pCASL 2025

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

pCASL 2525

PET Task	*PET Task*
Bed	*pCASL 2525*
Series Description	*pCASL 2525*
SCAN TIMING	
Number of Echoes	*1*
Receiver Bandwidth	*62.50*
Post Label Delay	*2525.0*
IMAGE ENHANCE	
Filter Choice	*None*
GATING/TRIGGER	
Pause After Navigator Prescan	*0*
FMRI	
PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*
SAT	
Tag Type	*None*
Fat/Water Saturation	*Fat*
TRICKS	
Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*
OTHERS	
Protocol Notes	*ASL Considerations: BOTTOM OF SLAB SHOULD BE AT BOTTOM OF CEREBELLUM. SUBJECTS EYES SHOULD BE OPEN. DO NOT OBLIQUE* *ASL Scan Parameter Tradeoffs:*

IMAGING PARAMETERS	
Imaging Mode	*3D*
Pulse Sequence	*3DASL*
Imaging Options	*EDR, Fast, Spiral*
SCAN RANGE	
FOV	*24.0*
Slice Thickness	*4.00*
Overlap Locations	*0*
Number of Slices	*1*
ACQ TIMING	
Freq	*768*
Phase	*6*
Freq DIR	*R/L*
NEX	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Quadrature (CP)*
Excitation Mode	*Selective*
USER CVS	
User CV2	*1.00*
User CV3	*1.00*
User CV4	*1.00*
User CV5	*1800.00*
User CV Mask2	*0*
MULTI-PHASE	
Seperate Series	*0*
Delay after Acquisition without AV	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*
DIFFUSION	
Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*
CONTRAST	
Contrast Yes/No	*No*

pCASL 2525

Property of Wallace Tumor Institute

Protocol: adult_head_PET/MR: BRAIN C11-PIB Alzheimer's V2 4/1/26

pCASL 3025

PET Task	*PET Task*
Bed	*pCASL 3025*
Series Description	*pCASL 3025*

SCAN TIMING

Number of Echoes	*1*
Receiver Bandwidth	*62.50*
Post Label Delay	*3025.0*

IMAGE ENHANCE

Filter Choice	*None*

GATING/TRIGGER

Pause After Navigator Prescan	*0*

FMRI

PSD Trigger	*Internal*
View Order	*Bottom/Up*
# of Repetitions REST	*0*
# of Repetitions ACTIVE	*0*

SAT

Tag Type	*None*
Fat/Water Saturation	*Fat*

TRICKS

Pause On/Off	*On*
Auto Subtract	*0*
Auto SCIC	*Off*

OTHERS

Protocol Notes	*ASL Considerations: BOTTOM OF SLAB SHOULD BE AT BOTTOM OF CEREBELLUM. SUBJECTS EYES SHOULD BE OPEN. DO NOT OBLIQUE* *ASL Scan Parameter Tradeoffs:*

IMAGING PARAMETERS

Imaging Mode	*3D*
Pulse Sequence	*3DASL*
Imaging Options	*EDR, Fast, Spiral*

SCAN RANGE

FOV	*24.0*
Slice Thickness	*4.00*
Overlap Locations	*0*
Number of Slices	*1*

ACQ TIMING

Freq	*768*
Phase	*6*
Freq DIR	*R/L*
NEX	*1.00*
Auto Shim	*Auto*
Phase Correction	*No*
RF Drive Mode	*Quadrature (CP)*
Excitation Mode	*Selective*

USER CVS

User CV2	*1.00*
User CV3	*1.00*
User CV4	*1.00*
User CV5	*1800.00*
User CV Mask2	*0*

MULTI-PHASE

Seperate Series	*0*
Delay after Acquisition without AV	*0*
Mask Phase	*0*
Mask Pause	*0*
Preserve	*0*

DIFFUSION

Recon All Images	*On*
# Synthetic b-values	*1*
Synthetic b-value	*1000.0;*

CONTRAST

Contrast Yes/No	*No*

pCASL 3025

Property of Wallace Tumor Institute